



Khawaja Yunus Ali University

Assignment

Name of the Department: Computer Science and Engineering

Course Code: CSE 0713-1103

Course Title: Electrical Circuit

Assignment No.: 01

Name of the Assignment : Calculation of Electricity bill of household .

Date of Submission : 05-02-2025

.....
Instructor Signature & Date

Submitted by –

Submitted to –

Name: Md Sojib Ahmed

ID Number: 06224205101006

Batch No: 18th

Semester: 1st Year 2nd Semester

Khawaja Yunus Ali University

Name: Sakil Ahammed

Lecturer

Department of EEE

Khawaja Yunus Ali University

Topic: Calculation of Electricity bill of household.

Step 1: The number of various equipment's available in my household.

- i) Light: 3
- ii) Refrigerator: 1
- iii) computer : 1
- iv) Water Pump : 1

Step 2: The power ratings of every equipment in my household.

- i) Light: 10w (each)
- ii) Refrigerator : 120 W
- iii) computer: 200W
- iv) Water Pump : 400W

Step 3: For each equipment, estimate the amount of time used.

- 1. Light: 7 hour per day.
- 2. Water Pump : 40 minute par day. = 0.667 hours
- 3. Refrigerator: : Chargers for 4 hours daily.
- 4. computer: 2 hour per day.

Step 4: calculate total energy consumed (in unit)

$$\text{Energy (kwh)} = \frac{\text{Power (W)} \times \text{Time (h)} \times \text{number}}{1000}$$

For, 3 light

$$= \frac{3 \times 7 \times 10}{1000} \times 30$$
$$= 7.2 \text{ unit}$$

For, 1 refrigerator

$$= \frac{1 \times 120 \times 6}{1000} \times 30$$
$$= 14.4 \text{ unit}$$

For, 1 computer

$$= \frac{1 \times 2 \times 200}{1000} \times 30$$
$$= 12 \text{ unit}$$

For ,1 water pump

$$= \frac{1 \times 0.667 \times 400}{1000} \times 30$$
$$= 8 \text{ unit}$$

Total energy consumed = (7.2+14.4+12+8) = 41.6 unit

Step 5:

0-75 unit = 4.63 taka

75-200 unit = 6.85 taka

Total cost of unit and other costs- Total cost of unit=(41.6× 4.63) =
192.60 taka

Meter rent = 10 taka

Demand charge = 40 taka

Vat(5%)= (192.60 × 5%) = 9.63 taka

Step 6:

Total electricity bill-

Total electricity bill of august = (192.61+10+40+9.63)=252.23 taka

Step 7:

Compare this bill with the bill of paper- Electricity bill of January on paper of bill = 249 taka

Compare = $(252.23 - 249) = 3.23$ taka

Step 8:

The reason for variation-

1. Estimation Errors: The estimated usage times or power ratings might not match the actual usage.
2. Standby Power Consumption: Many devices consume power even when turned off, which might not have been included.
3. Tariff Variations: The actual billing might include additional charges or different rates that weren't considered in the calculation.
4. Meter Accuracy: Differences between the household's meter readings and the utility company's recorded consumption could cause discrepancies



RECEIPT

Bill Information

Organization Name	Palli Bidyut (Postpaid)
Bill Month	January 2025
Bill Account Number	1009100912120
Bill Amount	BDT 249.0

Payment Information

bKash Account	01754301132
Payment Date	27 January, 2025
Transaction ID	CAR0ASPDYU
Paid to Organization	BDT 249.0
bKash Fee	0.0
Payment Received	249.00



This receipt has been generated
electronically